

Concept: *Relativistic time dilation*—Moving clocks run slow, compared to clocks at rest.

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - \frac{v^2}{c^2}}}$$

In the equation above, $\Delta t'$ is the time interval as perceived in the moving frame of reference, Δt is the time interval as perceived in the rest frame (not moving), v is the speed of the moving reference frame relative to the rest frame, and c is the speed of light: $c = 2.998 \times 10^8$ m/s.

You will receive 5 points for an on time submission. The final 20 points are assigned below. In order to get full credit for the problems below, be sure to:

- explicitly state all assumptions made that are not given in the problem.
- show all of the steps in each calculation.
- include proper physical units for all results.

Problem 1. (20 points) Imagine human space flight from Earth to Mars in the following two stages.

1. (4 pts) Go to NASA's website and look up information about the speed of the International Space Station (ISS) in its orbit around the Earth. Express this speed in units of meters per second.
2. (8 pts) Use Einstein's equation for time dilation to calculate how much slower clocks on the ISS run, compared to clocks at rest on the surface of the Earth. You can do this by calculating the ratio $\frac{\Delta t'}{\Delta t}$ (time elapsed on the ISS divided by time elapsed on Earth). **Note:** The answer should come out very close to 1, but the goal of this problem is to discover how different from 1 it really is. Be sure to do this with a computer or a calculator that can show enough significant digits. If you round off to 1, you will not receive points for this part.
3. (8 pts) Suppose one of the astronauts on the ISS has an identical twin of exactly the same age. If the astronaut spends a full year on the ISS (i.e., 1 year as experienced in the moving frame of reference of the ISS), how much older will the twin be who remained on Earth the whole time?

Problem 1. (20 points) Imagine human space flight from Earth to Mars in the following two stages.

1. (4 pts) Go to NASA's website and look up information about the speed of the International Space Station (ISS) in its orbit around the Earth. Express this speed in units of meters per second.
2. Answer (2pts for trying; 2 pts for right answer at order of magnitude; -2 for no source): <https://www.nasa.gov/international-space-station/space-station-facts-and-figures/>: 5 miles per second converts to 8047 m/s
3. (8 pts) Use Einstein's equation for time dilation to calculate how much slower clocks on the ISS run, compared to clocks at rest on the surface of the Earth. You can do this by calculating the ratio $\frac{\Delta t'}{\Delta t}$ (time elapsed on the ISS divided by time elapsed on Earth). **Note:** The answer should come out very close to 1, but the goal of this problem is to discovery how different from 1 it really is. Be sure to do this with a computer or a calculator that can show enough significant digits. If you round off to 1, you will not receive points for this part.
4. Answer (2 pts for trying; 4 pts for correct algebraic equation; 2 pts for correct number plug in.):

$$\frac{\Delta t'}{\Delta t} = \sqrt{1 - \frac{v^2}{c^2}} = \sqrt{1 - (8047\text{m/s})/(3 \times 10^8\text{m/s})} = 0.99998660491$$

5. (8 pts) Suppose one of the astronauts on the ISS has an identical twin of exactly the same age. If the astronaut spends a full year on the ISS (i.e., 1 year as experienced in the moving frame of reference of the ISS), how much older will the twin be who remained on Earth the whole time?
6. Answer (2 pts for trying; 4 pts for correct algebraic equation; 2 pts for correct number plug in.):

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1 \text{ yr}}{\sqrt{1 - (8047\text{m/s})/(3 \times 10^8\text{m/s})}} = 1\text{yr}/0.99998660491 = 1.00001339527\text{yr}$$

So the twin is $0.00001339527 \text{ yr} = 422.43323472 \text{ seconds}$ older.